

Motor Symptom Patterns and Severity in Parkinson's Disease

A Multivariate Analysis of Cognitive and Age-Related Associations

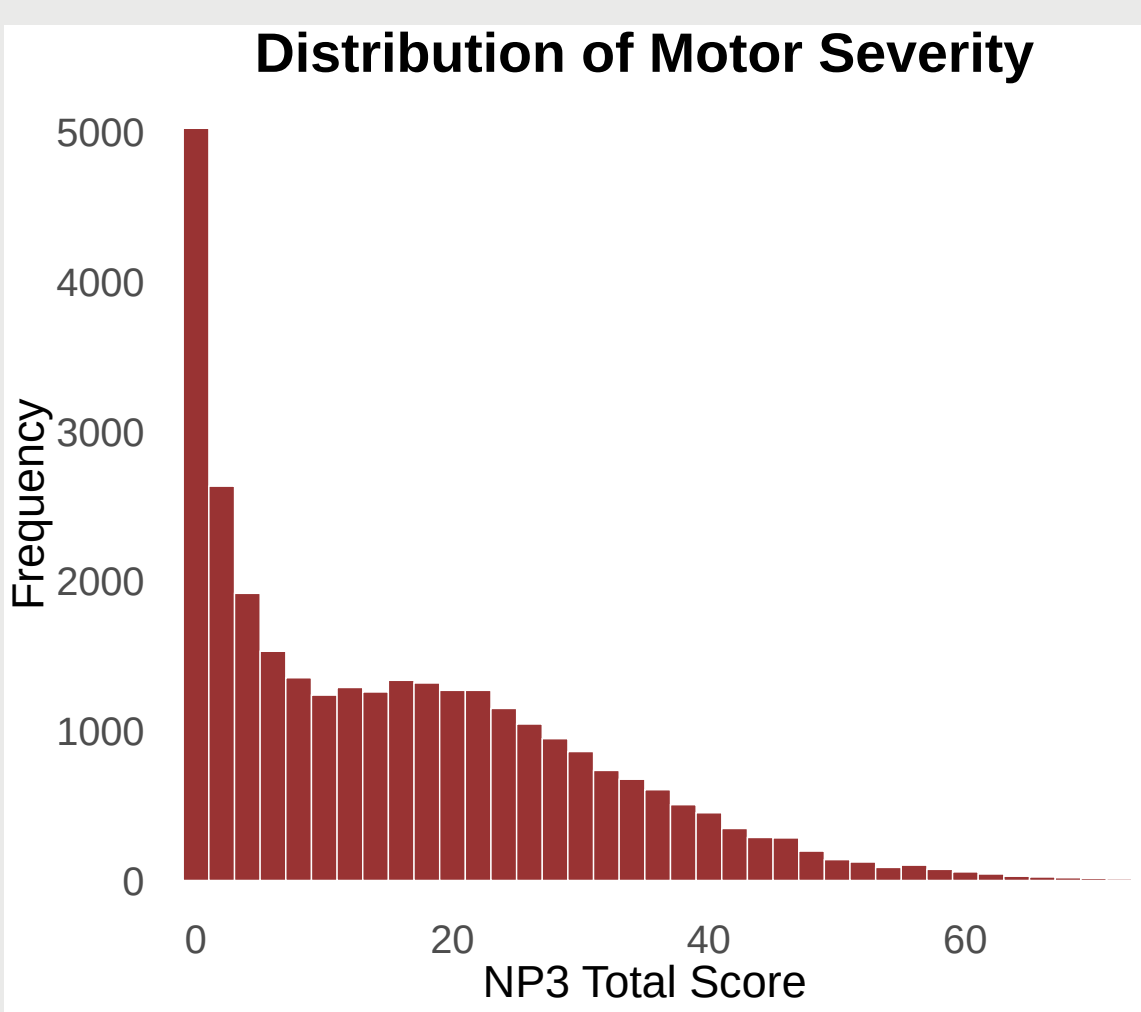


Introduction

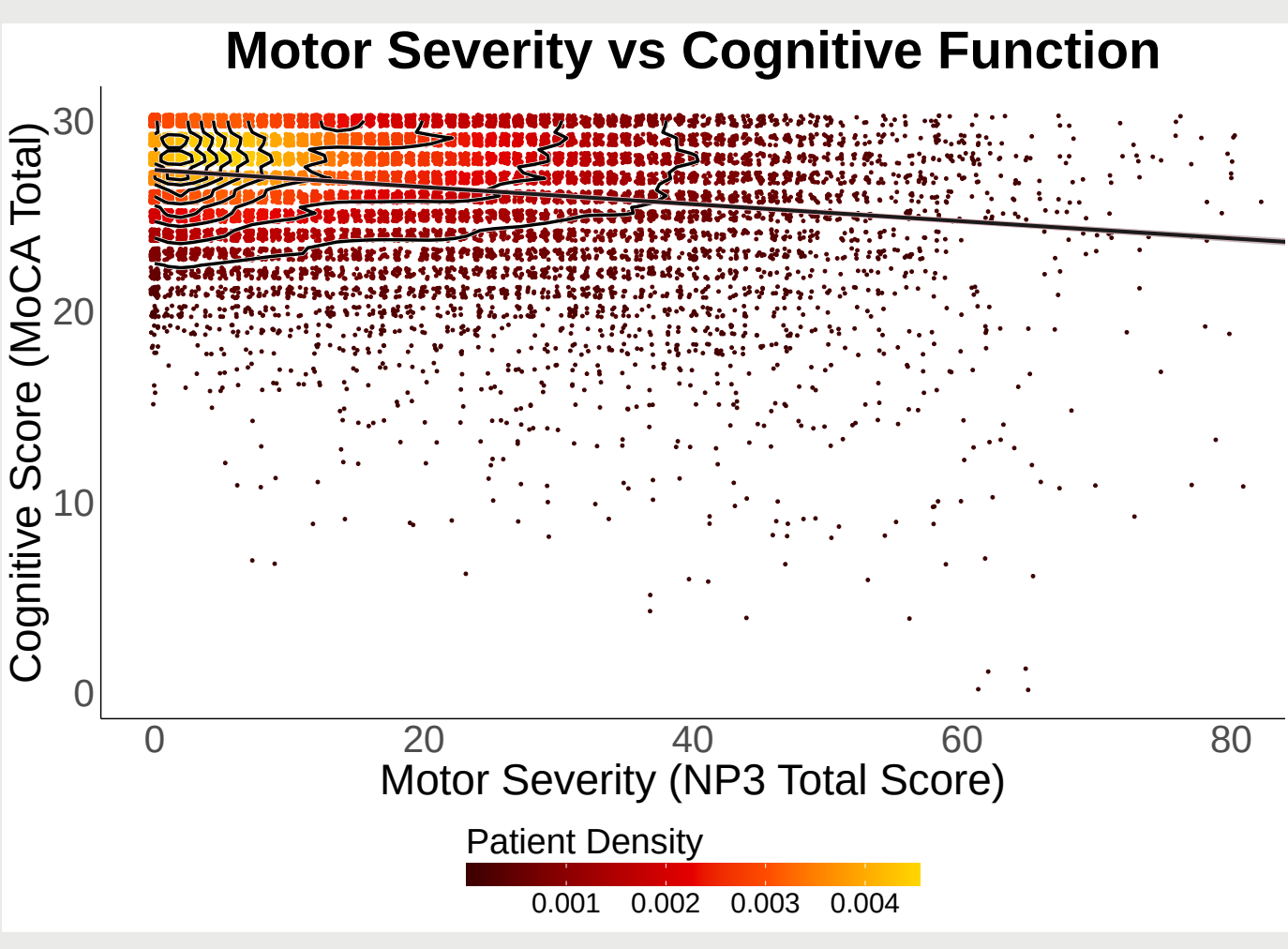
Parkinson's disease is a progressive neurodegenerative disorder characterized by motor impairments measured using the MDS-UPDRS Part III (NP3) scale. Understanding how motor severity relates to cognitive function and demographic factors may improve patient stratification and clinical management. However, the underlying structure and relationships among motor symptoms remain complex. This study aims to explore motor severity, symptom patterns, and their associations with cognition and age using statistical and multivariate techniques.

Methods

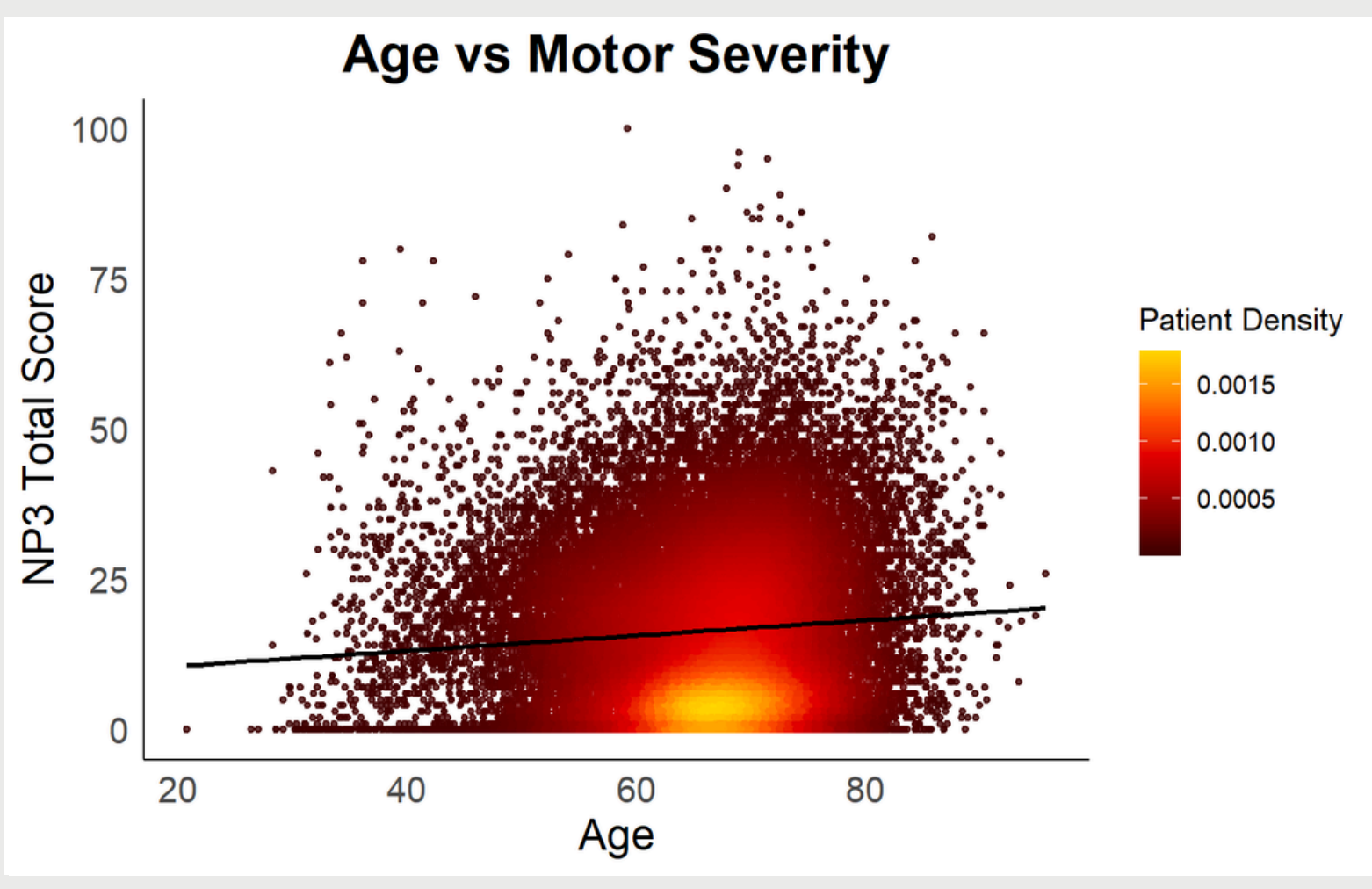
- Cohort: approx 4,900 participants; clinical scores for MDS-UPDRS Part III (NP3) and MoCA.
- Distributions: Binned histograms to characterize motor severity and population skew.
- Trends: Heat-scatter plots quantifying associations between motor severity, age, and cognition.
- Correlations: Spearman heatmaps to map co-occurrence among motor sub-scores.
- Phenotyping: PCA to visualize clinical biplots and distinct motor clusters.



Distribution of NP3 total scores (bin width = 2), showing a concentration of participants at lower severity levels.

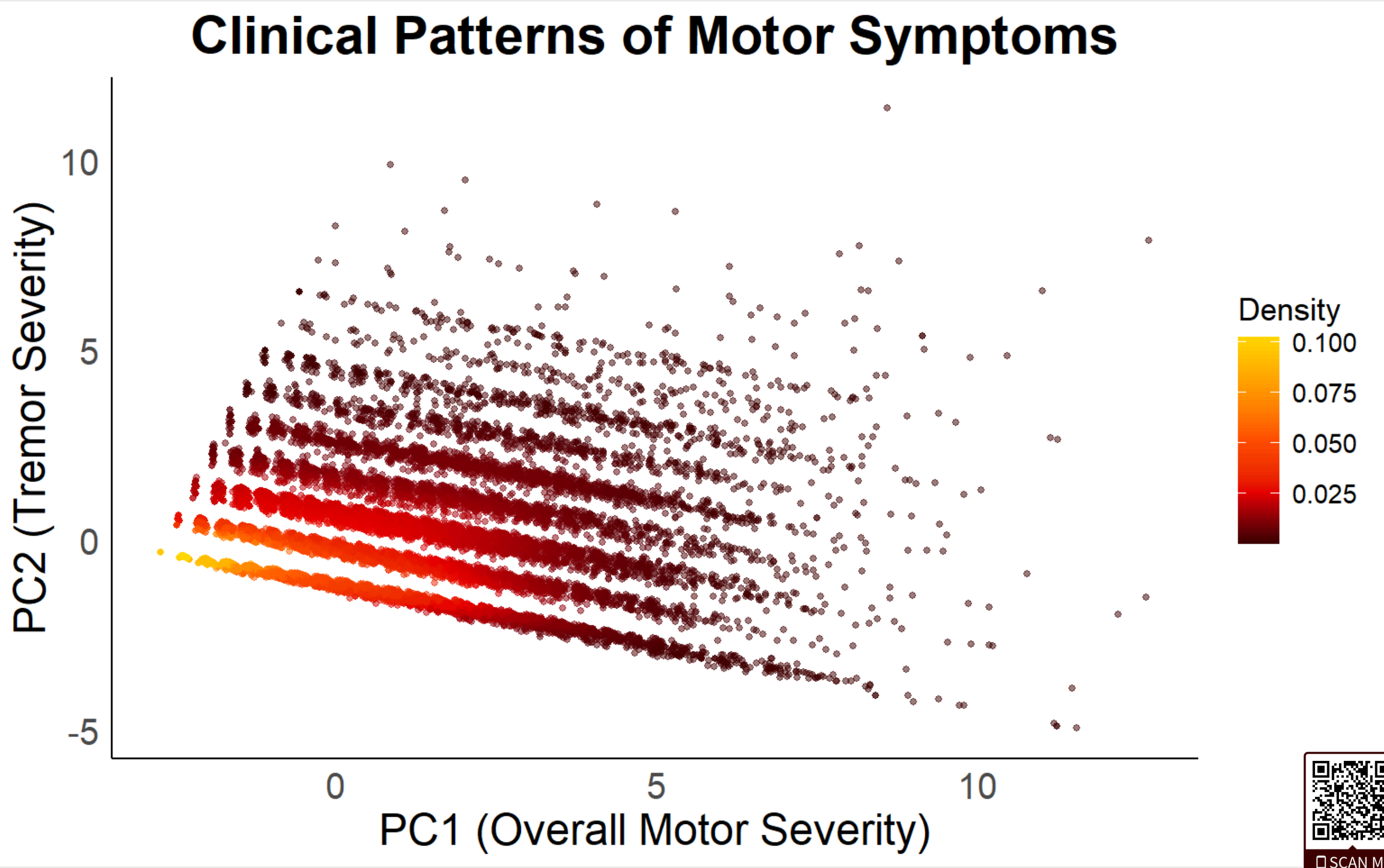


Relationship between motor severity (NP3) and cognitive function (MoCA), with point density indicating participant concentration.

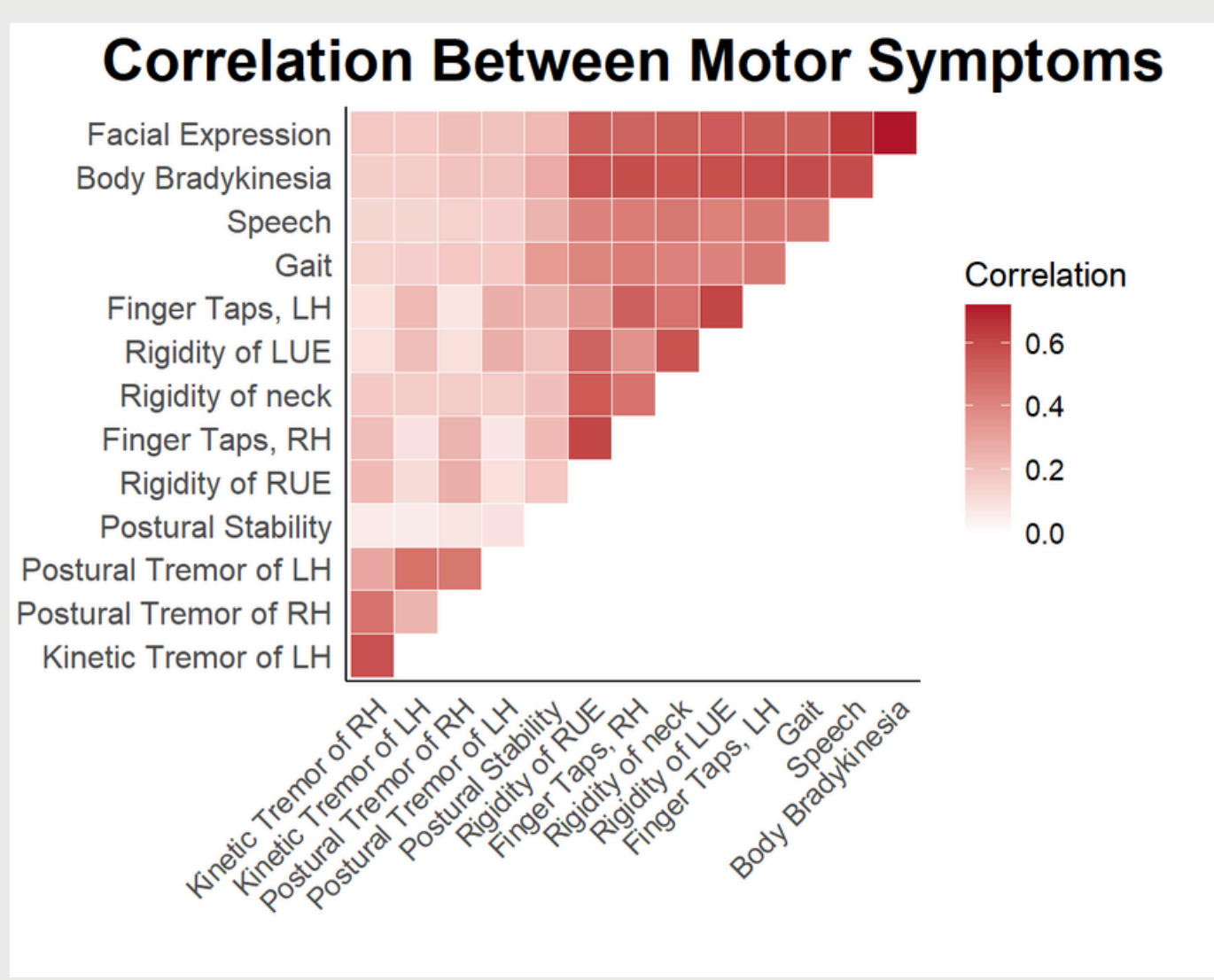


Relationship between age and motor severity (NP3), with density highlighting the distribution of participants across age groups.

Principal Component Analysis of motor symptoms. PC1 represents overall motor severity, and PC2 represents tremor-related variation.



Spearman correlation matrix of motor symptom sub-scores, where darker colors indicate stronger positive correlations. (L = left, R = right, H = hand, U = upper, E = extremity)



Conclusion

Motor severity in this cohort is generally mild and not strongly explained by age or cognitive decline. The observed clustering of bradykinesia-related symptoms and divergence between tremor and non-tremor patterns suggest distinct motor subtypes, stressing the importance of accounting for heterogeneity in disease progression.

References

- Parkinson Progression Markers Initiative (PPMI). Progress Neurobiology, 2011
- Marek K et al. Annals of Clinical and Translational Neurology, 2018
- Goetz CG et al. Movement Disorders, 2008
- Fahn S, Elton RL. Recent Developments in Parkinson's Disease, 1987

Key Insights

- Motor severity is right-skewed, with most participants showing mild impairment
- Weak negative association between motor severity and cognitive function
- Weak positive association between age and motor severity
- Gait, speech, and facial expression form a highly synchronized cluster, suggesting these symptoms manifest/worsen at a similar rate.
- The "fanning" pattern confirms that as disease severity (PC1) increases, patients diverge into two distinct groups: those with prominent tremors and those whose impairment is dominated by non-tremor symptoms.